

Biomechanical Comparison of Quadriceps Tendon Fixation With Patellar Tendon Bone Plug Interference Fixation in Cruciate Ligament Reconstruction

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Purpose: The purpose of this study was to use current fixation techniques and compare the stiffness and ultimate tensile failure of the tendinous end of the quadriceps tendon (QT) with the bone plug end of the bone-patellar tendon-bone (BPTB) graft using current techniques of fixation. **Type of Study:** Randomized trial of elderly cadaver knees. **Materials and Methods:** Tibial and femoral biodegradable interference fixation and femoral EndoButton (Smith & Nephew, Acufex, Mansfield, MA) fixation in bone tunnels with the QT and the BPTB graft were compared by using 10 pairs of elderly cadavers and biomechanical testing. Two groups, fixation at time zero (simulating fixation in the operating room) and testing after 1,000 loading cycles (simulating patient rehabilitation exercises), were used. **Results:** At time zero fixation, stiffness of the soft tissue QT tibial tunnel interference fixation was 59% less stiff than the stiffness of the interference fixation of a BPTB plug in a femoral tunnel ($P = .11$). The EndoButton femoral fixation resulted in a decrease in stiffness at time zero compared with femoral tunnel interference fixation of the soft tissue QT ($P = .03$). All groups improved stiffness with cycling the construct to 1,000 cycles. **Conclusions:** Placement of the QT tendinous end of the graft in the femoral bone tunnel when using a interference fixation will approximate the stiffness of a bone plug in the tibial bone tunnel with interference fixation. The EndoButton fixation is not as stiff as either of the femoral interference fixation options. The addition of more than 20 loading cycles could remove laxity from the graft fixation-graft cruciate ligament complex and improve its stiffness. **Key Words:** Quadriceps tendon—Patellar tendon—Cruciate ligament fixation.

The patellar tendon autograft has been associated with significant donor-site morbidity. Anterior knee pain, pain on kneeling, patellar tendon rupture,

and fat pad herniation can occur after graft harvest. The quadrupled hamstring autograft donor-site morbidity may include anterior cruciate ligament (ACL) agonist weakness and disruption of the protective ACL proprioceptive arc. For these reasons as well as the purpose of revision surgery or knees with an ACL and posterior cruciate ligament (PCL) deficiency, the quadriceps tendon (QT) has been proposed as a graft alternative for cruciate ligament reconstruction.¹⁻³ The QT offers excellent biomechanical strength, large cross-section area, appropriate length, and a bone plug that can be harvested from the patella.^{4,5} The QT can be harvested by a second surgeon to shorten operating time.³ The tendon is divided into several lamina that can be split into 2 separate graft tails, making it an option in techniques that use 2 bundles to reconstruct the ACL or the PCL. However, the tendon end of the quadriceps graft requires soft-tissue fixation of the

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graft in a bone tunnel. Conventional soft-tissue fixation methods, including the EndoButton (Smith & Nephew, Acufex, Mansfield, MA) with tied polyester tape, are less stiff than interference fixation of a bone plug. Interference screw fixation against a tendon graft has improved stiffness but resulted in lower failure loads.⁶

Poly(lactic acid) (PLA) or its copolymers fabricated as an interference screw may possess several advantages over a metal interference screw (the gold standard of graft fixation), such as drill-through capabilities in revision surgery, compatibility with magnetic resonance imaging, and multitasking. Biodegradable interference screw fixation in a bone plug model appears equal in biomechanical testing to a metal interference screw,^{7,8} and it may be superior in load to failure in a soft-tissue model to a metal interference screw.^{9,10} A direct comparison of soft-tissue biodegradable interference screw fixation of QT with biodegradable interference screw of a bone plug has not been performed at present. Optimum fixation of the QT would provide equal stiffness and load at failure with both the soft tissue and bone plug ends of the graft. Graft fixation with present rehabilitation programs is subject to repetitive cyclic loads that may cause knee laxity due to graft slippage secondary to insecure fixation.

The purpose of this study was to compare the stiffness and ultimate tensile load-bearing abilities of the EndoButton (tied with polyester tape) and interference fixation of the soft-tissue end of the QT both at time zero (simulating fixation in the operating room) and testing after 1,000 loading cycles (simulating patient rehabilitation exercises). Ultimate failure, stiffness, and displacement to failure of the tendinous end of the QT were compared using these fixation devices with biodegradable interference fixation of the bone plug end of the bone–patellar tendon–bone (BPTB) graft.

MATERIALS AND METHODS

Arthritis-free, paired cadaver knees were used for this study. In group A, which represented fixation at time zero, 5 paired knees (10 total knees), mean age 59.9 years (range, 40 to 77 years), were prepared as will be described and tested to load to failure. In group B, 5 paired knees (10 total knees), mean age 62.3 years (range, 46 to 81 years), represented the cyclic loading group. This cyclic loading group was prepared, cycled 1,000 times at 1 Hz from 10 to 110 N for 1,000 cycles, and then loaded to failure. A maximum

load of 110 N was chosen to avoid catastrophic failure during the cyclic loading protocol.

Bone mineral density (BMD) was measured in each femur and tibia with dual photon absorptiometry (DEXA) scanning system (Hologic QDR-1000 Whole body X-Ray Bone Densitometer; Hologic Inc, Waltham, MA) after removing soft tissue from the femur and tibia. The patellar tendon, QT, and the patella were left attached to the tibia. All soft tissue was stripped from the femur. The femur and tibia were placed in a saline bath during the scanning process. The measurements were taken in the metaphyseal area of the tibia and femora at the location of the interference screw fixation.

Two grafts, a patellar tendon and a QT, were harvested from each knee. One of the grafts was tested in the femoral tunnel and the other graft was tested in the tibial tunnel. The full-thickness QT, 8 cm long, was harvested with a bone plug from the patella, 10 mm wide and 25 mm long. The BPTB grafts, 10 mm wide, with bone plugs 10 mm wide and 25 mm long, were harvested from each knee. The QT and patellar tendon bone plugs were harvested side by side from the patella. Random number table was used to assign which graft was placed in the tibia and which graft was tested on the femur of a given knee (Table 1). This allowed for the side of the cadaver and the bone, femur and tibia, to be controlled in the comparison. Also, the soft-tissue fixation (QT) could be directly compared with the BPTB plug fixation in both the femur and tibia of each pair of knees.

For the QT group, the tendon end of the graft was measured with custom graft sleeves in 0.5-mm incre-

TABLE 1. The Experimental Setup

	Femur		Tibia	
	Right	Left	Right	Left
Testing after 20 cycles				
98-00305	BPTB	QT	QT	BPTB
98-03004	QT	BPTB	BPTB	QT
98-03006	QT	BPTB	BPTB	QT
98-02001	BPTB	QT	QT	BPTB
98-6003	BPTB	QT	QT	BPTB
Testing after 1,000 cycles				
98-6002	QT	BPTB	BPTB	QT
98-9896	BPTB	QT	QT	BPTB
98-6012	QT	BPTB	BPTB	QT
100002	BPTB	QT	QT	BPTB
11003	QT	BPTB	BPTB	QT

NOTE. BPTB and QT position in the bone tunnel were determined by random number assignment.

ments. This was performed in a sequential process, starting with a sleeve diameter that was obviously too large and then by using progressively smaller diameters until the smallest possible dimension was achieved. The tunnels were drilled with reamers whose diameters were increased in 0.5-mm increments, corresponding to the graft sleeves. For example, a 9.5-mm graft was placed in a 9.5-mm tunnel. This tendon was trimmed to approximately 10 mm (mean, 9.85 mm) in diameter. The bone plugs were sized to a 10-mm tunnel. Both groups A and B were prepared in a similar manner.

The tunnels were placed anatomically on the tibia and the femur for ACL reconstruction. The tibial tunnel entrance was midway between the tibial tubercle and the posterior medial edge of the proximal tibia, approximately 3 finger breadths below the joint line. The exit for the tibial tunnel was the posterior medial footprint of the native ACL. The femoral socket entrance was adjacent to the over-the-top and at approximately the 1:30/10:30 position on the face of an imaginary clock. The guide wire for the socket was directed proximally to replicate the endoscopic femoral tunnel placement. The femoral cortex was drilled with the 4.5-mm EndoButton reamer to allow graft passage and EndoButton fixation in applicable femurs. The 25-mm length BioScrew (Linvatec, Largo, FL) was placed as interference fixation directly against the QT end (soft tissue) of the graft and as close to joint surfaces as possible. Insertion torque was measured in Newton-millimeters, as the screw was placed anterior to the graft. Seven-millimeter screws were used in the femoral tunnels and 9-mm screws were used in the tibial tunnels. The patellar tendon bone plug was fixed with the BioScrew in the same fashion as the QT. The femoral QT grafts were prepared with an EndoButton, and No. 5 cottony polyester tape was woven through the end of the tendon after the fashion of a Kessler stitch (Fig 1). The bone plug of the QT and the opposite end of the patella tendon graft were secured in autobody filler (Bondo) in 1- and ½-inch polyvinyl chloride pipe (Fig 2), which was placed in a custom-made grip attached to an Instron materials testing system (Instron, Canton, MA). This technique limited grip slippage, because fixation did not rely on soft tissue.

Specimens were mounted in a Servohydraulic testing machine (Instron Model 1331 with an 8500+ controller) using a custom-made jig (Fig 3). The tibia or the femur with the graft and fixation construct was preloaded with a 10-N force. Group A, "fixation at time zero," was pretensioned to 10 N and cycled at 1

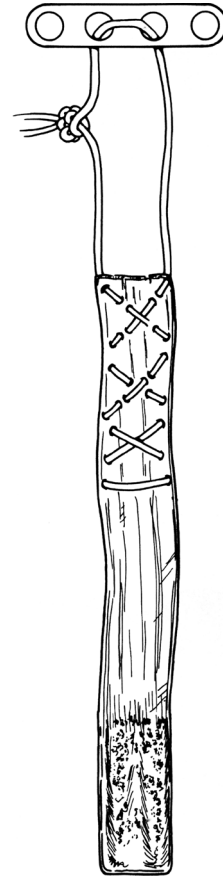


FIGURE 1. The QT with No. 5 polyester tape linking the tendon graft to the EndoButton. The No. 5 polyester tape is woven through the tendon end after the fashion of the Kessler.

Hz from 10 to 50 N for 20 cycles to replicate the cycling that is performed in the operating room before the tibial portion of the graft is fixed. Group B, the "rehabilitative" group, was pretensioned to 10 N and cycled at 1 Hz from 10 to 110 N for 1,000 cycles.¹¹ The maximum value of 110 N was chosen to avoid graft fixation failure while cycling and to represent presumed low loads of rehabilitation. Specimens were positioned so that the direction of pull was parallel to the axis of graft insertion in the bone tunnel, or in line with the graft and bone tunnel. Grafts were loaded with a 60 mm/min tensile force to a pullout failure load determined by a significant deviation in linearity of the load-deformation curve (which indicates graft-tunnel separation). Graft displacement from the bone tunnel was measured by the Instron cross-head movement. Mechanical testing data were acquired and analyzed using Instron Series IX software and a personal computer that was online with the Servohydraulic

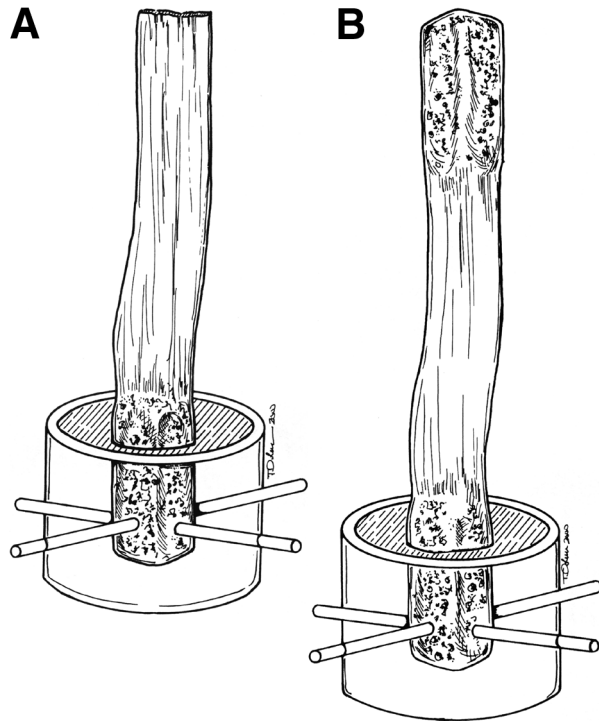


FIGURE 2. Both the (A) QT and the (B) BPTB grafts had a bone plug potted with Bondo with K-wire used as reinforcement rods in PVC pipe. This construct was used to decrease grip slippage.

device. Statistical comparisons for each fixation type were preformed for BMD (g/cm^2), torque (N-mm), maximum tensile pullout load (N), stiffness (N/mm), and displacement to failure (mm) by using an analysis of variance (Statview software; SAS Institute, Cary, NC). The Scheffe post hoc criteria were used, and $P < .05$ was considered indicative of significant differences.

RESULTS

Group A Fixation at Time Zero

Interference Fixation: There was no difference in ultimate failure load among the 4 groups of interference fixation (Fig 4). Stiffness varied from 61.8 N/mm for femoral tunnel bone plug fixation to a low of 36.2 N/mm for a soft-tissue end of the QT in a tibial tunnel (Table 2), but these differences were not significant ($P = .11$). In the 5 knees that had femoral tunnel bone plug fixation and tibial tunnel soft-tissue QT fixation, 3 had a difference in stiffness of greater than 20 N/mm. In the 5 knees that had femoral soft-tissue QT and tibial bone plug fixation, none had a difference in stiffness of greater than 20 N/mm. Tibial tunnel bone

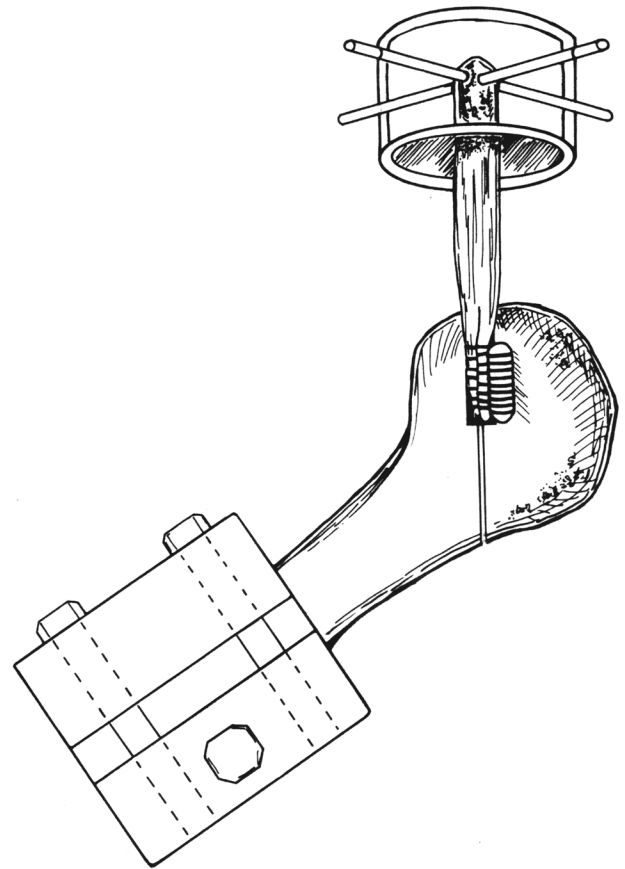


FIGURE 3. The tested bone, either the tibia or the femur, is held in a custom-made grip. The graft is secured in the bone tunnel with the fixation that is to be tested. The potted bone plug fits in a custom-made grip that attaches to the Instron load cell.

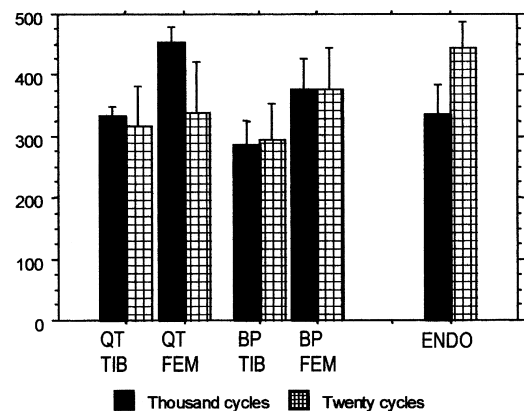


FIGURE 4. The results of ultimate failure in Newtons showed no significant differences among the groups in this study. QT and bone plug (BP) fixation under cyclic loading (1,000 cycles) and initial fixation (20 cycles) were compared in tibial (TIB) and femoral (FEM) tunnels. The EndoButton (ENDO) was compared in femoral tunnel under both of the cycling protocols. The error bars represent 1 standard deviation.

TABLE 2. Results of Biodegradable Interference Fixation Ultimate Failure and Stiffness on Testing at Time Zero

	Femur		Tibia	
	BPTB	QT	BPTB	QT
Testing after 20 cycles				
Stiffness N/mm	61.8 (20.0)	53.7 (14.6)	45.4 (15.3)	36.2 (9.5)
Ultimate failure N	377 (150)	339 (185)	293 (137)	318 (146)
Failure mode	BP pullout	QT slipped	BP pullout	QT slipped

NOTE. Standard deviation is expressed in parentheses.

plug interference fixation was slightly, but not significantly, stiffer than the stiffness of the QT soft-tissue interference fixation in a tibial tunnel. In all cases, the mode of failure was graft slippage around the biodegradable screw, except for 1 case of patellar tendon rupture at the transition zone into the bone block.

EndoButton: Although the tensile failure load was larger for the EndoButton tied with polyester tape, it was not significantly different from the femoral tunnel interference screw fixation of the soft tissue QT (Table 3). However, stiffness and particularly displacement before failure were statistically significantly worse with the EndoButton ($P = .03$) (Table 3). Displacements before failure were up to 3 times greater than interference fixation ($P = .01$). The mode of failure was stretch of the polyester tape and, in 1 case, the EndoButton pulling through bone.

Group B Rehabilitative Loads

Interference Fixation: There was residual laxity in grafts and graft fixation that was removed with cycling. Much of this stress relaxation occurred in the first 100 cycles. This resulted in significant improvements in stiffness and displacement before failure in the QT soft-tissue interference constructs. Improvement also occurred with femoral tunnel bone plug

fixation but the increase was not significant (Tables 4, 5, and 6) After 1,000 cycles, the stretch was removed, particularly from the soft-tissue QT fixation. Thus, the levels of ultimate failure, stiffness, and displacement before failure were similar among all interference screw groups after 1,000 cycles (Table 6). Two femoral tunnel soft-tissue QT constructs, 2 tibial tunnel soft-tissue QT constructs, and 1 tibial tunnel bone plug construct failed with cycling, all before 100 cycles were completed. All femurs and tibias in which this occurred had a BMD of less than 0.6 g/cm² (Table 7). However, not all bones with a BMD of less than 0.6 g/cm² failed on cycling. Graft slippage past the interference screws was the mechanism for all interference fixation specimens.

EndoButton: Stiffness and displacement before failure improved with cycling to 1,000 cycles using the EndoButton for fixation. The EndoButton displaced more at failure than the femoral tunnel bone plug fixation ($P = .04$) but not significantly more than the displacement of the QT fixed with interference screw ($P = .13$). Stiffness of the EndoButton was less but not significantly less than either of the femoral tunnel interference fixation constructs, the bone plug ($P = .16$) and the soft-tissue QT ($P = .11$). The mechanism of failure was stretch of the linkage tape.

Group A Fixation at Time Zero and Group B Rehabilitative Loads

All fixation methods showed comparable levels of maximum tensile load-bearing capabilities. This included all interference screw groups and the EndoButton groups (Fig 4). Group B, which was cycled 1,000 times, did not alter the load at failure. The loads to failure were lower than those that have been reported using young cadaver bone.^{12,13}

DISCUSSION

Based on this biomechanical study, we recommend placing the tendinous end of the QT in the femoral

TABLE 3. Comparison of Testing at Time Zero, Group A, of Femoral Fixation of a QT

Construct	Ultimate Failure (N)	Displacement (mm)	Stiffness (N/mm)	Mode of Failure
Femoral QT	339 (185)	11.2 (1.9)	53.7 (14.6)	Slippage of tendon from the bone tunnel
EndoButton with Mersilene tape (QT)	445.2 (93.1)	27.5 (8.0)	24.4 (5.0)	Stretch of the Mersilene tape
<i>P</i> value	.5478	.0008	.0258	

NOTE. Biodegradable interference fixation of QT and EndoButton fixation of a QT in all parameters measured is compared in this table. Standard deviation is expressed in parentheses. Statistical significance to a level of $P < .05$ is identified in bold.

TABLE 4. Comparison of Displacement and Stiffness Among Femoral Fixation Options in Group A and Group B

	Femoral PT Bone Plug		Femoral QT		QT With EndoButton	
	D (mm)	K (N/mm)	D (mm)	K (N/mm)	D (mm)	K (N/mm)
20 cycles	8.3 (1.9)	61.8 (20.0)	11.2 (1.9)	53.7 (14.6)	27.5 (8.0)	24.4 (5.0)
1,000 cycles	6.2 (1.6)	86.7 (23.2)	7.1 (1.4)	94.8 (20.8)	13.4 (5.7)	59.1 (18)
P value	.1035	.1062	.0206	.0158	.0125	.0038

NOTE. Standard deviation is expressed in parentheses. Statistical significance to a level of $P < .05$ is identified in bold. Displacement (D) and stiffness (K) are compared with 20 cycles and 1,000 cycles.

tunnel to approximate the stiffness of the bone plug in a tibial tunnel. Tibial tunnel soft-tissue fixation is not as stiff as femoral tunnel bone plug fixation in fixation at time zero. The stiffness of femoral tunnel soft-tissue interference fixation is a closer approximation to tibial tunnel bone plug interference fixation. A mismatch of stiffness will allow 1 side of the graft construct to be loaded unequally, even if each end of the graft has a similar load at failure. We cycled the grafts in the first portion of the study for 20 cycles to approximate our clinical protocol in the operating room. After the graft is fixed on the femur, the knee is cycled 20 to 25 times from 0° to 110° to stress the graft and femoral fixation. Group A is a better reflection of the clinical situation at time zero.

Within the fixation options tested in the rehabilitative group, there was no difference in load to failure, stiffness, and displacement at failure between the interference screw fixation groups. If one compares femoral tunnel interference fixation in these parameters, there was no difference between a bone plug and tendon fixation. Similarly, tibial interference fixation was not different between a bone plug and tendon fixation.

Six of 10 QT constructs and 9 of 10 bone plug constructs survived 1,000 cycles and were tested to failure (Table 7). All cycling failures were in bones,

both femurs and tibias, that had poor BMD, less than 0.6 g/cm^2 , which may not be seen in the clinical situation. With 2 pairs of knees with both tibias and femurs of BMD less than 0.6 g/cm^2 , all QT soft-tissue interference fixation screw failed catastrophically with cycling. These same 2 pairs of knees had the only bone plug catastrophic failure with cycling. In a given knee, if the femur BMD was greater than 0.6 g/cm^2 , catastrophic failure did not occur with cycling in this study. Table 8 compares our cadaveric BMD with a clinical study of Vuori et al.¹⁴ who evaluated metaphyseal BMD in young female patients in a fashion similar to the method we used. In our study, BMD was correlated with ultimate failure in the interference screw groups. Catastrophic failure at low cyclic loads may not occur with better bone quality, but it cannot be excluded based on our data.

In the first portion of the study, the EndoButton had significantly greater displacement to failure and less stiffness than the other 2 femoral tunnel constructs, the QT soft-tissue interference screw and the bone

TABLE 5. Comparison of Displacement and Stiffness Among Tibial Interference Fixation Groups After 20 Cycles and 1,000 Cycles

	Tibial PT Bone Plug		Tibial QT	
	D (mm)	K (N/mm)	D (mm)	K (N/mm)
20 cycles	8.6 (1.8)	45.4 (15.3)	11.7 (3.5)	36.2 (9.5)
1,000 cycles	4.6 (1.2)	96.9 (11.1)	5.9 (0.7)	92 (1.0)
P value	.0067	.0008	.0350	.0001

NOTE. Standard deviation is expressed in parentheses. Statistical significance to a level of $P < .05$ is identified in bold. Displacement (D) and stiffness (K) are compared with 20 cycles and 1,000 cycles.

TABLE 6. Comparative Analysis of Load to Failure and Stiffness With Interference Fixation of Bone Plug and QT in Both Test Groups After 20 Cycles and 1,000 Cycles

	Femur		Tibia	
	BPTB	QT	BPTB	QT
Testing after 20 cycles				
Stiffness N/mm	61.8 (20.0)	53.7 (14.6)	45.4 (15.3)	36.2 (9.5)
Ultimate failure N	377 (150)	339 (185)	293 (137)	318 (146)
Testing after 1,000 cycles				
Stiffness N/mm	86.7 (23.2)	94.8 (20.8)	96.9 (11.1)	92.0 (1.0)
Ultimate failure N	379 (107)	454 (44)	287 (78)	334 (26)

NOTE. Standard deviation is expressed in parentheses.

TABLE 7. *The Number of Specimens in Each Group Tested to Failure After the Cycling Regimens*

	Femur		Tibia	
	BPTB	QT	BPTB	QT
Static testing 20 cycles	N = 5	N = 5	N = 5	N = 5
1,000 cycles	N = 5	N = 3	N = 4	N = 3
Failures during cycling	N = 0	N = 2	N = 1	N = 2

NOTE. The bottom row are the specimens that did not survive 1,000 cycles.

plug interference screw. The EndoButton graft with the linkage system used in our study will displace at least twice as much at failure as the QT soft-tissue interference screw. Movement through the linkage of the graft with the EndoButton has been noted in another investigation.¹⁵ This longitudinal graft motion has been labeled the "bungee cord effect."¹⁶ Indeed, our results support this concept. However, in the second portion of the study, the EndoButton displacement and stiffness improved with cyclic loading and were comparable to the other fixation options in the rehabilitative group.

All constructs showed residual laxity after 20 cycles, which was group A (Tables 4 and 5). It has been estimated that 1,000 cycles represents 1 week of rehabilitation from ACL reconstructive surgery.¹¹ The cycling that occurs with rehabilitation likely creates laxity in the graft construct. Either improved fixation for bone plug and soft tissue or a means to remove laxity in the graft and fixation may improve the results of reconstruction. Displacement at failure and stiffness reflect laxity with a load in the cruciate ligament graft and graft fixation. Cycling initial fixation in the operating room could be of value in removing laxity from the graft fixation-graft cruciate ligament complex and improve stiffness. A device to cycle the graft and femoral fixation through the tibial tunnel in endoscopic techniques based on this work would im-

prove the biomechanical properties, displacement, and stiffness of the construct. The inferior stiffness of present fixation options may be a source of graft laxity as rehabilitation progresses. Combining fixation devices, i.e., the use of hybrid fixation techniques, may improve both the stiffness and the ultimate tensile load-bearing abilities of the technique and should be explored in future research.

CONCLUSIONS

1. Among all groups tested, there was no significant difference in load to failure.
2. Tendinous interference fixation of the QT in the tibial tunnel was less stiff than femoral tunnel BPTB fixation at time zero. The tendon end of the QT at this time should be placed in the femoral tunnel when using biodegradable interference fixation.
3. Femoral tunnel tendon interference fixation with a biodegradable interference screw was comparable to femoral tunnel BPTB interference fixation in stiffness, displacement to failure, and ultimate failure, both with 20 and 1,000 cycles. Similarly, tibial tunnel interference fixation either with a BPTB or QT was comparable in group B.
4. The EndoButton had significantly more displacement at failure and less stiffness than QT biodegradable interference fixation at time zero.
5. In all groups except the femoral BPTB option, 1,000 cycles improved the displacement at failure and stiffness to a level of statistical significance. This suggests that further cycling in the operating room beyond 20 cycles of the graft and graft fixation would improve the biomechanical characteristics of stiffness and displacement and load to failure.

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TABLE 8. *The BMD of Cadavers in the Present Study Were Compared With Similar Area of the Tibial and Femoral Metaphysis in Young Female Patients of Vuori et al.¹⁴*

	Tibia g/cm ²	Femur g/cm ²
Cadavers in present study	0.485	0.693
	SD, 0.167	SD, 0.162
Young female patients	1.090	1.283
Vuori et al.	SE, 0.053	SE, 0.183

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